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(54) SPEAKER APPARATUS

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(57) **ABSTRACT**

An electronic apparatus may include a main housing configured to accommodate electronic components and including an elastic hole, and a sub-housing configured to be coupled to the main housing through the elastic hole. The main housing may include a deformation portion provided along one side of the elastic hole, a body portion provided along another side of the elastic hole, and a connection rib provided across the elastic hole to connect and support the deformation portion and the body portion. The sub-housing may include a hook inserted into the elastic hole to couple the sub-housing to the main housing, and during coupling of the sub-housing to the main housing, the deformation portion may undergo elastic deformation to enable the hook to be inserted into the elastic hole.

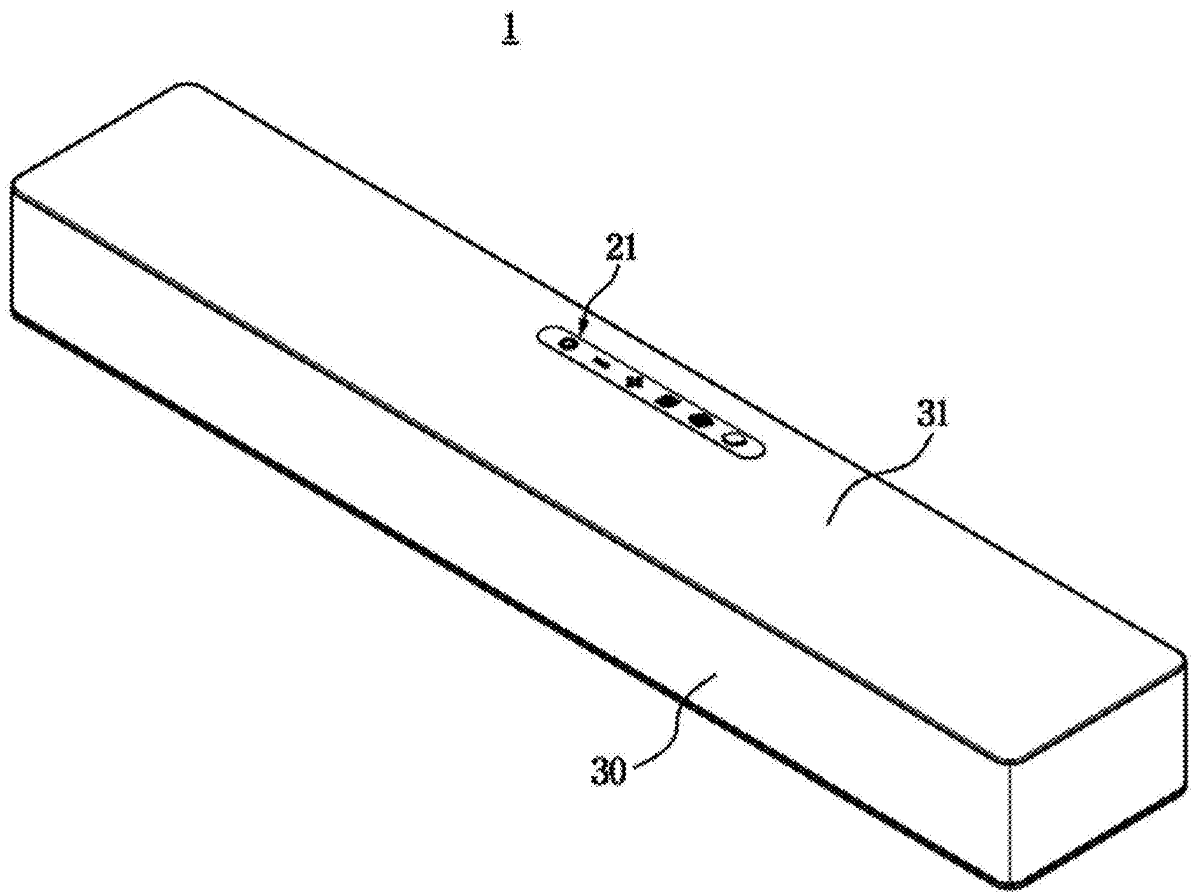


FIG. 1

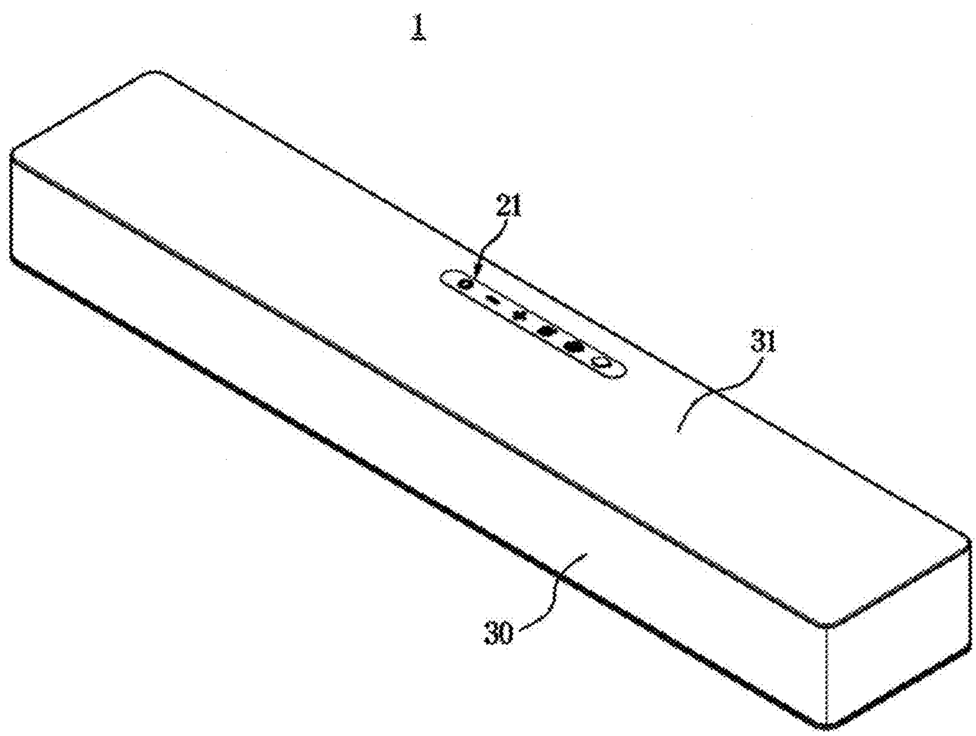


FIG. 2

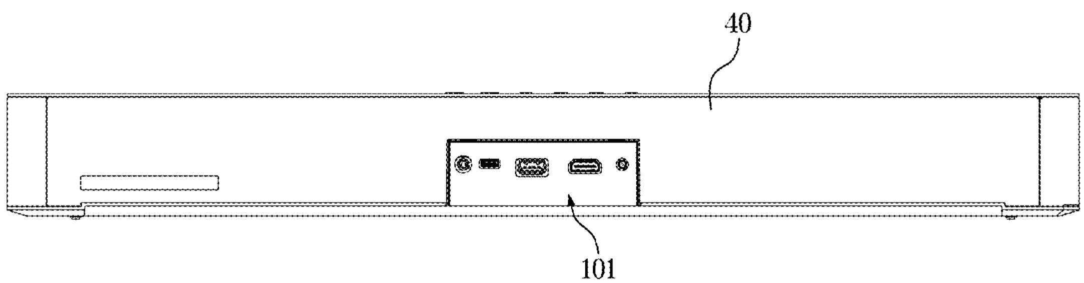
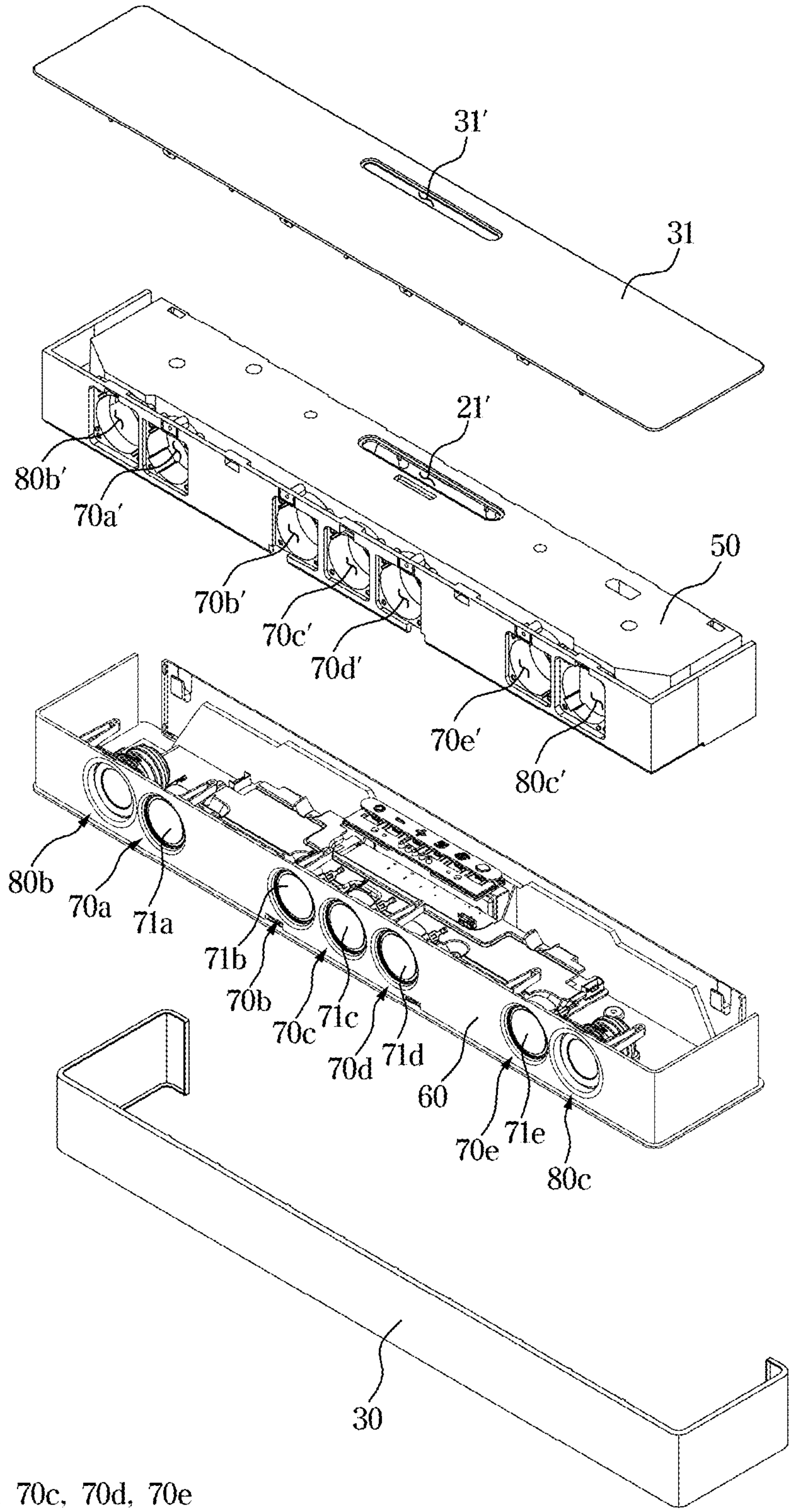
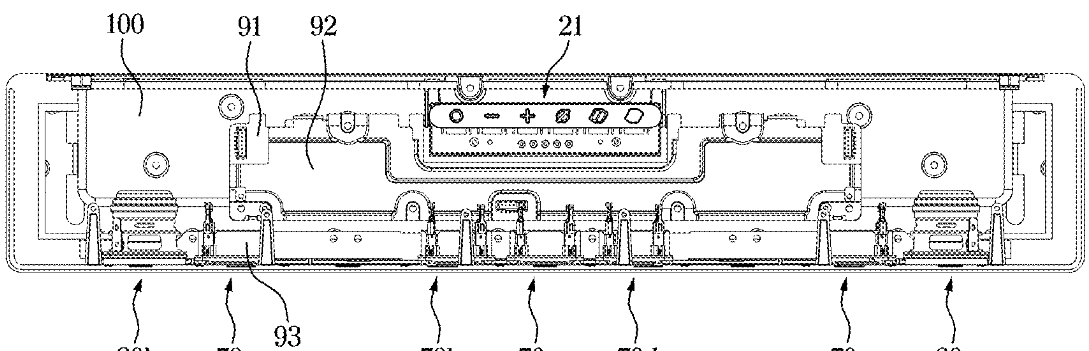


FIG. 3



70: 70a, 70b, 70c, 70d, 70e
71: 71a, 71b, 71c, 71d, 71e

FIG. 4



70: 70a, 70b, 70c, 70d, 70e

FIG. 5

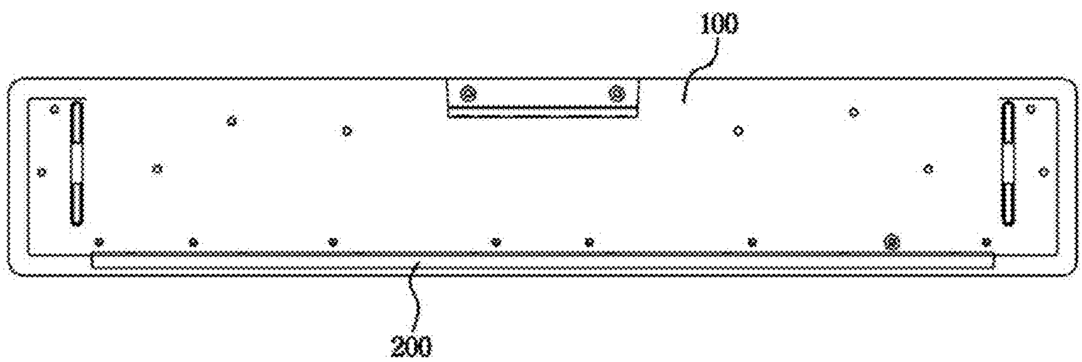


FIG. 6

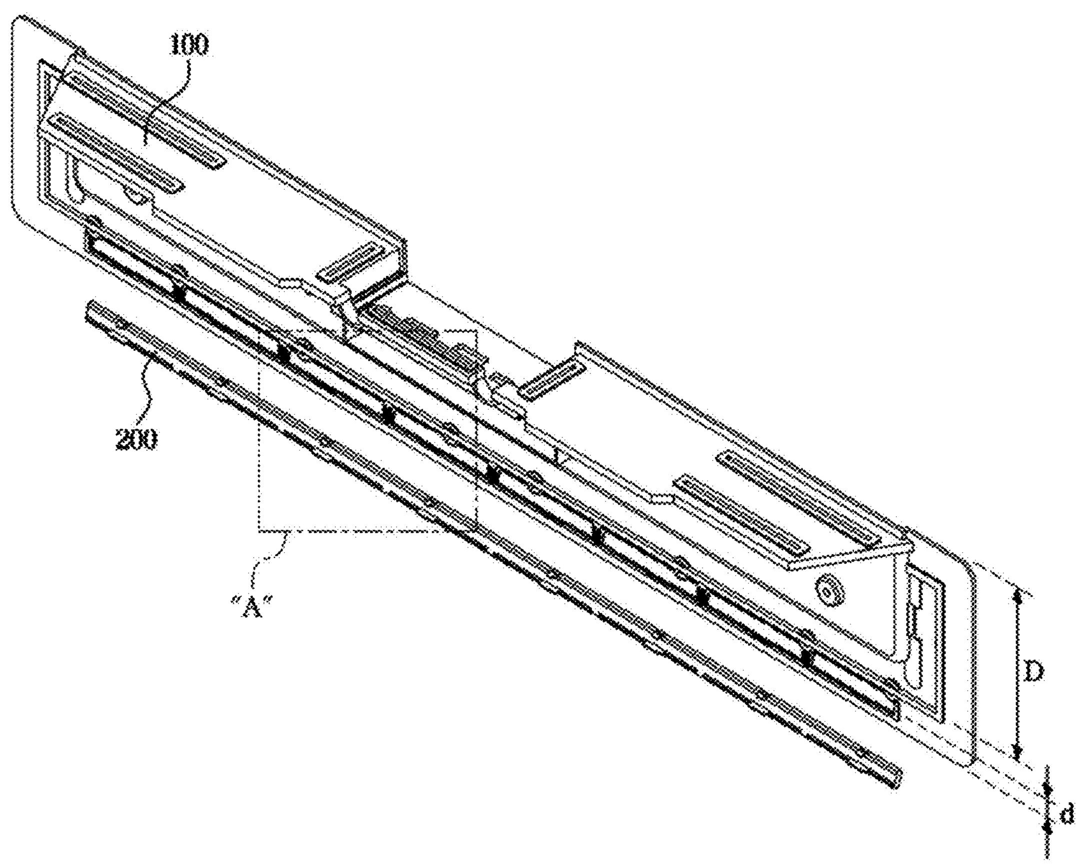
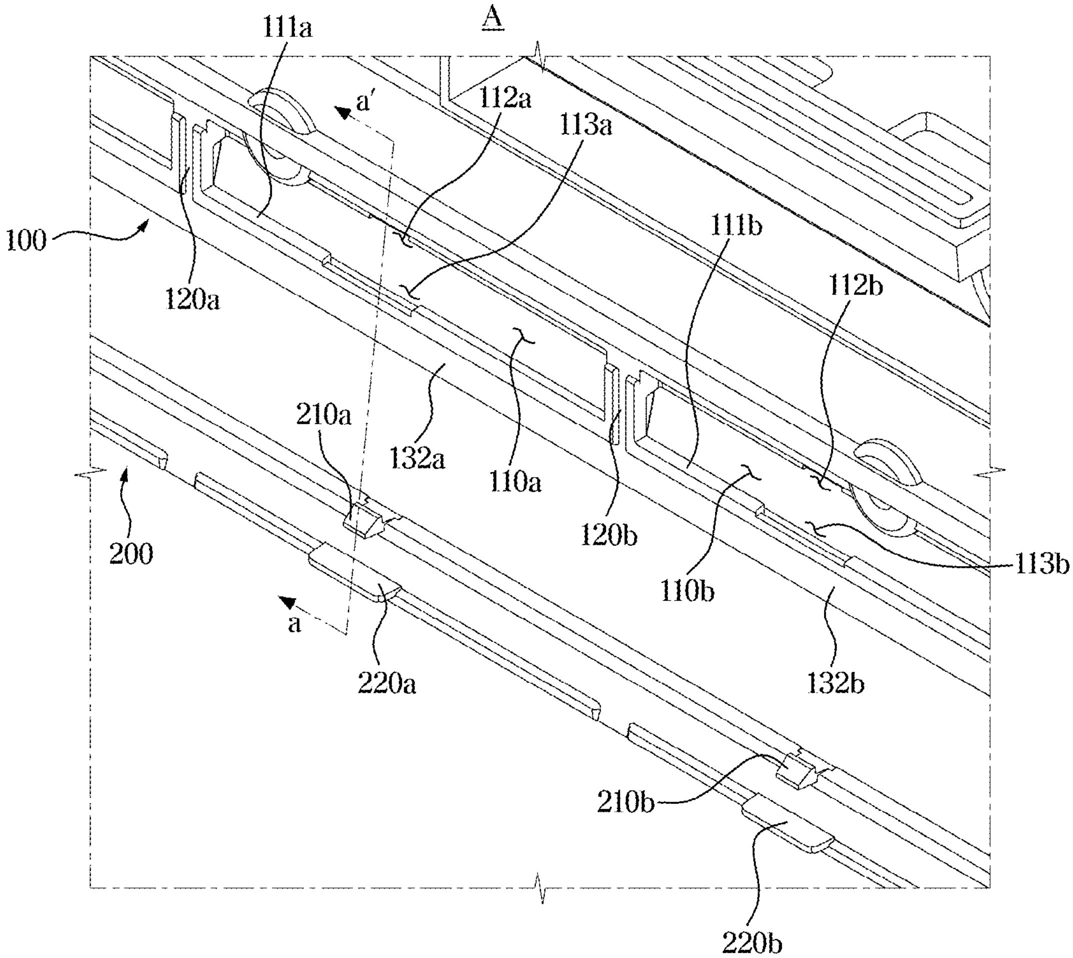


FIG. 7



110: 110a, 110b
111: 111a, 111b
112: 112a, 112b
113: 113a, 113b
120: 120a, 120b
132: 132a, 132b
220: 220a, 220b

FIG. 8

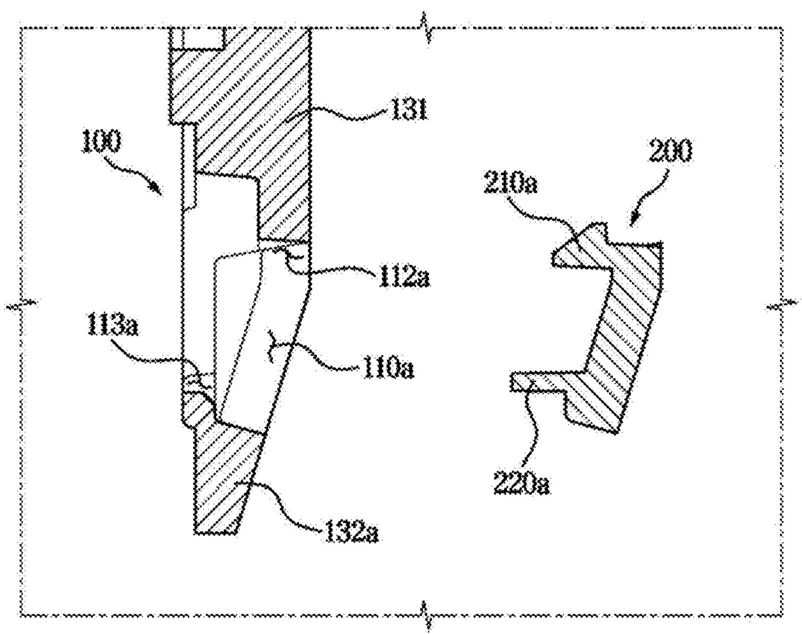


FIG. 9

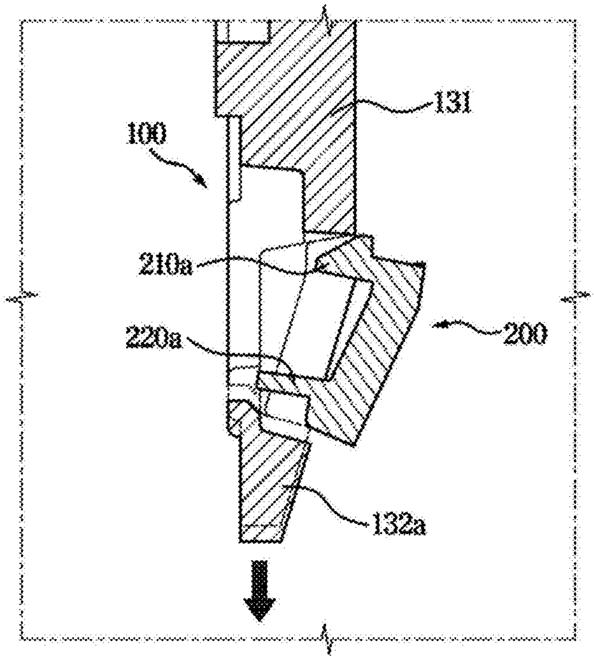


FIG. 10

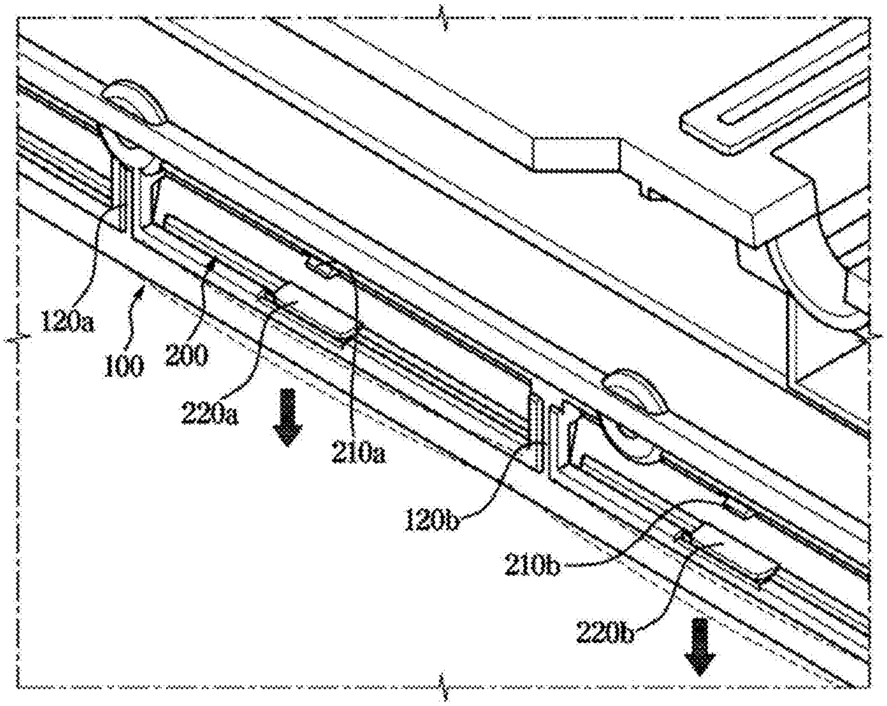


FIG. 11

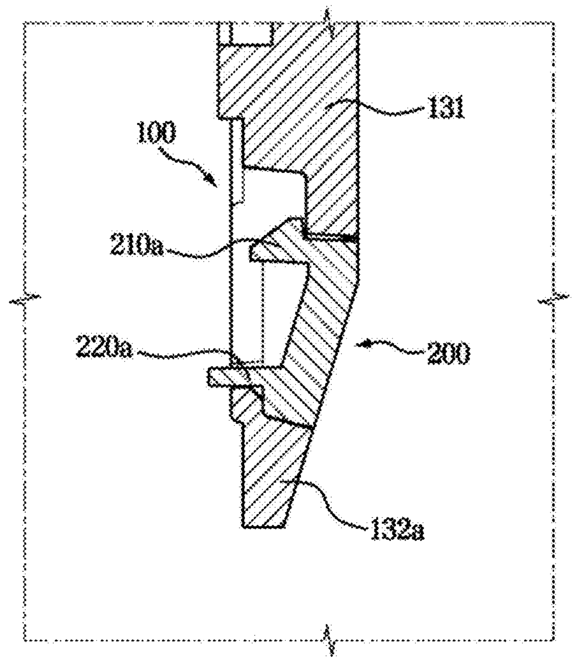
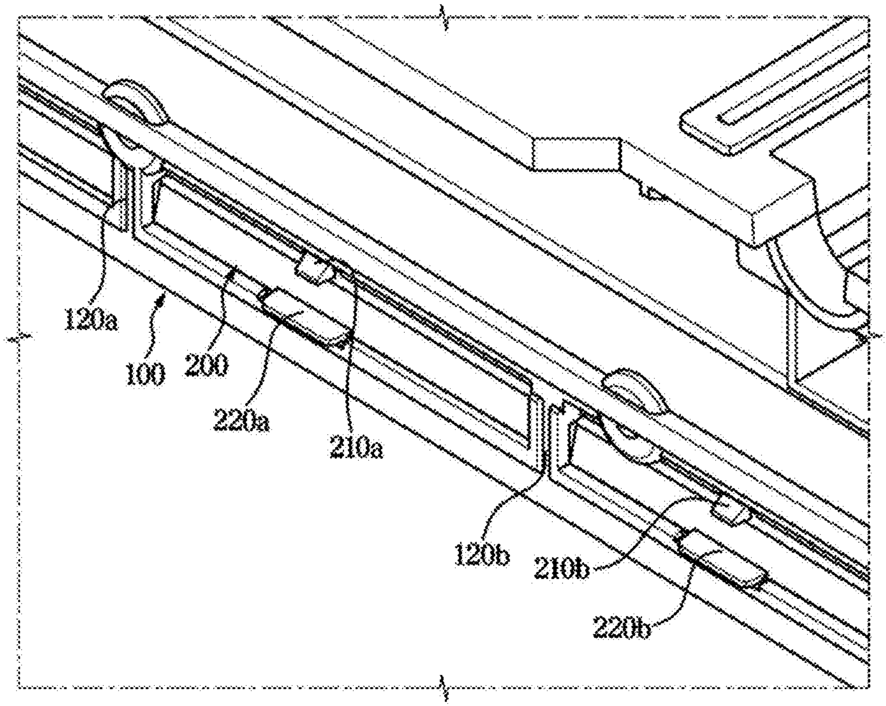


FIG. 12



SPEAKER APPARATUS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a bypass continuation application of International Patent Application No. PCT/KR2023/017381, filed on Nov. 2, 2023, which claims priority to Korean Patent Application No. 10-2023-0000939, filed on Jan. 3, 2023, in the Korean Intellectual Property Office, the disclosure of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] Apparatuses and methods consistent with example embodiments relate to a speaker apparatus.

[0003] In general, a speaker is a device that generates sound by converting electrical signals into vibrations of a vibrating member.

[0004] A speaker may include a moving system, a suspension system, and a magnet system. The moving system may include components that are directly involved in sound reproduction, such as a diaphragm and a voice coil. The suspension system may include components, such as a damper and an edge, that support the outer and inner parts of the diaphragm in the correct position. The magnet system may refer to a system to optimize the movement of the diaphragm by allowing the magnetic field from the permanent magnets and the magnetic fields created by the voice coil to interact more efficiently.

[0005] Speakers may be categorized according to their shape as cone speakers, dome speakers, flat-panel speakers, ribbon speakers, horn speakers, and micro speakers.

[0006] As the shapes of speakers have become more diverse, the methods of designing the appearance of speakers have also become more diverse. In recent years, a design that installs a lamp inside the speaker and attaches a cover made of a light-transmissive material to some areas of the speaker's cover, allowing various colors of light generated by the lighting of the lamp to be transmitted to create an aesthetic effect, has become popular.

SUMMARY

[0007] One or more embodiments of the present disclosure provides a speaker apparatus including a coupling structure capable of reducing the length of a hook.

[0008] One or more embodiments of the present disclosure provides a speaker apparatus including a coupling structure capable of preventing damage to a hook.

[0009] One or more embodiments of the present disclosure provides a speaker apparatus capable of efficiently utilizing internal space.

[0010] An aspect of the present disclosure provides an electronic apparatus may include a main housing configured to accommodate electronic components and including an elastic hole, and a sub-housing configured to be coupled to the main housing through the elastic hole. The main housing may include a deformation portion provided along one side of the elastic hole, a body portion provided along another side of the elastic hole, and a connection rib provided across the elastic hole to connect and support the deformation portion and the body portion. The sub-housing may include a hook inserted into the elastic hole to couple the sub-housing to the main housing, and during coupling of the sub-housing to the main housing, the deformation portion may undergo elastic deformation to enable the hook to be inserted into the elastic hole.

[0011] The elastic hole may include a first elastic hole and a second elastic hole separated by the connection rib, and the hook may include a first hook inserted into the first elastic hole and a second hook inserted into the second elastic hole.

[0012] The deformation portion may include a first deformation portion arranged to correspond to the first elastic hole and a second deformation portion arranged to correspond to the second elastic hole. The first deformation portion and the second deformation portion may be configured to elastically deform, respectively, during the coupling of the sub-housing to the main housing.

[0013] The sub-housing may include a guide rib configured to guide reception of the sub-housing into the elastic hole, and the guide rib may be configured to press the deformation portion of the main housing during the coupling of the sub-housing to the main housing.

[0014] The main housing may include a side wall protruding from the deformation portion to define one side of the elastic hole, and a guide rib groove provided in the side wall to allow the guide rib to be seated therein.

[0015] The main housing may include a hook fastening groove to which the hook is fastened, and the hook fastening groove may be at a position corresponding to the guide rib groove.

[0016] The guide rib may extend in a direction corresponding to a direction in which the hook extends from the sub-housing.

[0017] The elastic hole and the deformation portion may extend along a longitudinal direction of the main housing.

[0018] The connection rib may include a plurality of connection ribs provided along a direction in which the elastic hole extends, and the plurality of connection ribs may be spaced apart from each other to partition the elastic hole.

[0019] The sub-housing may be coupled to the main housing to cover the elastic hole.

[0020] A width of the deformation portion may be narrower than a width of the body portion.

[0021] The sub-housing may include a light-transmissive material.

[0022] The electronic apparatus may be a speaker apparatus.

[0023] A speaker apparatus according to one or more embodiments of the present disclosure includes a speaker. The speaker apparatus includes an electronic component arranged to be in contact with the speaker. The speaker apparatus includes a main housing accommodating the speaker and the electronic component and including an elastic hole. The speaker apparatus includes a sub-housing coupled to the main housing to cover the elastic hole. The main housing includes a deformation portion arranged along one side of the elastic hole. The main housing further includes a body portion arranged along the other side of the elastic hole. The sub-housing further includes a hook inserted into the elastic hole to support the sub-housing to the main housing. The sub-housing further includes a guide rib configured to press against the deformation portion of the main housing when the sub-housing is coupled to the main housing. The deformation portion elastically deforms when the sub-housing is coupled to the main housing, causing the hook to be inserted into the elastic hole.

[0024] The main housing may further include a connection rib arranged across the elastic hole to connect and support the deformation portion with the body portion.

[0025] The elastic hole may include a first elastic hole divided by the connection rib. The elastic hole may include a second elastic hole divided by the connection rib. The hook may include a first hook inserted into the first elastic hole. The hook may include a second hook inserted into the second elastic hole.

[0026] The deformation portion may include a first deformation portion arranged to correspond to the first elastic hole. The deformation portion may include a second deformation portion arranged to correspond to the second elastic hole. The first deformation portion and the second deformation portion may be configured to elastically deform, respectively, when the sub-housing is coupled to the main housing.

[0027] The main housing may further include a side wall protruding from the deformation portion to define one side of the elastic hole. The main housing may further include a guide rib groove formed in the side wall to allow the guide rib to be seated therein.

[0028] The main housing may further include a hook fastening groove to which the hook is fastened. The hook fastening groove may be formed at a position corresponding to the guide rib groove.

[0029] The elastic hole and the deformation portion may extend along a longitudinal direction of the main housing. The connection rib may include a plurality of connection ribs arranged along a direction in which the elastic hole extends. The plurality of connection ribs may be arranged to be spaced apart from each other to partition the elastic hole.

[0030] The speaker apparatus may further include a light emitting lamp configured to emit light in electrical contact with the electronic component. The sub-housing may include a light-transmissive material such that light emitted by the light emitting lamp is transmitted.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. **1** is a view illustrating a speaker apparatus according to one or more embodiments.

[0032] FIG. **2** is a view illustrating the speaker apparatus of FIG. **1** from the rear.

[0033] FIG. **3** is an exploded view illustrating the speaker apparatus according to one or more embodiments.

[0034] FIG. **4** is a view illustrating an interior of the speaker apparatus according to one or more embodiments.

[0035] FIG. **5** is a view illustrating a state in which a sub-housing is fastened to a main housing according to one or more embodiments.

[0036] FIG. **6** is a view illustrating only the main housing and the sub-housing separated from the speaker apparatus according to one or more embodiments.

[0037] FIG. **7** is an enlarged view of area A of FIG. **6**.

[0038] FIG. **8** is a side cross-sectional view after cutting the main housing and the sub-housing of FIG. **7** along the cutting surface a-a'.

[0039] FIG. **9** is a view illustrating a state in which the sub-housing is fastened to the main housing as a deformation portion of FIG. **8** elastically deforms.

[0040] FIG. **10** is a view illustrating a state in which the sub-housing is fastened to the main housing as a first deformation portion and a second deformation portion according to one or more embodiments are elastically deformed, respectively.

[0041] FIG. **11** is a view illustrating a state in which the sub-housing is fastened to the main housing according to one or more embodiments.

[0042] FIG. **12** is a view of FIG. **11** from another angle.

DETAILED DESCRIPTION

[0043] Various embodiments of the present document and terms used therein are not intended to limit the technical features described in this document to specific embodiments, and should be understood to include various modifications, equivalents, or substitutes of the corresponding embodiments.

[0044] In addition, the same reference numerals or signs shown in the drawings of the disclosure indicate elements or components performing substantially the same function.

[0045] Also, the terms used herein are used to describe the embodiments and are not intended to limit and/or restrict the disclosure. The singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. In this disclosure, the terms "including", "having", and the like are used to specify features, figures, steps, operations, elements, components, or combinations thereof, but do not preclude the presence or addition of one or more of the features, figures, steps, operations, elements, components, or combinations thereof.

[0046] It will be understood that, although the terms "first", "second", "primary", "secondary", etc., may be used herein to describe various elements, but elements are not limited by these terms. These terms are only used to distinguish one element from another element. For example, without departing from the scope of the disclosure, a first element may be termed as a second element, and a second element may be termed as a first element. The term of "and/or" includes a plurality of combinations of relevant items or any one item among a plurality of relevant items.

[0047] Further, as used in the disclosure, the terms "front", "rear", "top", "bottom", "side", "left", "right", "upper", "lower", and the like are defined with reference to the drawings, and are not intended to limit the shape and position of any element.

[0048] When a given element is referred to as being "connected to", "coupled to", "supported by" or "in contact with" another element, it is to be understood that it may be directly or indirectly connected to, coupled to, supported by, or in contact with the other element. When a given element is indirectly connected to, coupled to, supported by, or in contact with another element, it is to be understood that it may be connected to, coupled to, supported by, or in contact with the other element through a third element.

[0049] It will also be understood that when an element is referred to as being "on" another element, it may be directly on the other element or intervening elements may also be present.

[0050] In referring to the direction of rotation, a clockwise direction may be referred to as a first direction and a counterclockwise direction, which is the opposite of the first direction, as a second direction. Such representations may be used in describing specific details for implementing the disclosure, but the direction of rotation of the elements of the disclosure is not limited by these terms.

[0051] One or more embodiments of the present disclosure provide a coupling structure that may be universally applicable to assemble decorative components along the edges of a product (e.g., a main housing of an electronic apparatus such as a speaker). Elastic deformation of a mating component (e.g., the main housing) may be used to achieve secure engagement. The coupling structure may be applicable at peripheral or border regions of the main housing, where both space constraints and aesthetic considerations are critical.

[0052] According to various embodiments of the present disclosure, the main housing may undergo elastic deformation to engage with the sub-housing, thereby eliminating the need for the hook on the sub-housing to include additional length for its own elastic deformation.

[0053] According to various embodiments of the present disclosure, the hook may remain rigid without elastic deformation, thereby preventing potential breakage of the hook during assembly. By shifting the elastic deformation requirement to the main housing, the hook does not need to be thin or highly flexible. This may result in greater mechanical strength and reduces the risk of hook fracture or failure, even under repeated use.

[0054] As the hook may not require additional length for elastic deformation, the internal space within the speaker apparatus may be utilized more efficiently, corresponding to the reduced hook length. Compared to conventional hook-type designs, embodiments of the present disclosure may achieve the same or better coupling performance with a height reduction of at least 5.0 mm, enabling use in thin or tightly packaged assemblies without spatial limitations.

[0055] Hereinafter, various embodiments according to the disclosure will be described in detail with reference to the accompanying drawings.

[0056] FIG. **1** is a view illustrating a speaker apparatus according to one or more embodiments. FIG. **2** is a view illustrating the speaker apparatus of FIG. **1** from the rear. FIG. **3** is an exploded view illustrating the speaker apparatus according to one or more embodiments.

[0057] Referring to FIGS. **1** to **3**, a speaker apparatus **1** may include a main speaker **70**. The main speaker **70** is a device that generates sound by converting electrical signals into vibrations of a vibration plate **71** (e.g., a diaphragm). The main speaker **70** may be provided in a plurality, and may be arranged symmetrically with respect to each other. More particularly, the main speakers **70** may be arranged in planar symmetry with respect to each other based on the center of an inner housing **50**. The main speaker apparatus **1** may be formed in a bar-type shape. However, the position in which the main speaker **70** is arranged and the shape of the speaker apparatus **1** are not limited thereto.

[0058] The main speaker **70** is configured to include a magnetic circuitry and a vibration system.

[0059] The magnetic circuitry may be a portion to which an electrical signal is transmitted to generate sound in the main speaker **70**. The magnetic circuitry is configured to include a magnet and a pole piece disposed in the center of the magnet.

[0060] The vibration system may be a portion in which sound is generated and output as vibration, and be configured to include a voice coil arranged in a gap formed between the magnet and the pole piece, a bobbin on which the voice coil is installed, and the vibration plate **71** movably arranged by a magnetic circuit in which the magnetic circuitry is formed.

[0061] With such a structure, the main speaker **70** may convert electrical signals into vibrations of the vibration plate **71**, and may generate sound by generating compressional waves in the air through the vibrations. The operating principle of the main speaker **70** described above may be equally applied to the operation of front speakers **80***b* and **80***c*, which will be described below.

[0062] The speaker apparatus **1** may include the inner housing **50** on which components of the speaker apparatus **1**, such as the speaker **70** are mounted, and a plurality of covers **30** and **40** configured to cover the inner housing **50**. Components of the speaker apparatus **1** may be arranged and mounted on the inner housing **50**. The inner housing **50** will be described in more detail later.

[0063] The plurality of covers **30** and **40** may be arranged to cover the inner housing **50**, and may be configured to protect the components of the speaker apparatus **1** that are mounted within the inner housing **50**. The plurality of covers **30** and **40** will be described in more detail later.

[0064] The speaker apparatus **1** may include an upper cover **31** configured to cover the inner housing **50**. The upper cover **31** may be mounted on an upper portion of the inner housing **50**. The upper cover **31** may include a grid pattern. Thereby, the aesthetics of the speaker apparatus **1** may be improved.

[0065] The speaker apparatus **1** may include a rear cover **40** configured to cover the inner housing **50**. The rear cover **40** may be mounted on a rear side of the inner housing **50** to protect the components that are disposed within the inner housing **50**.

[0066] The speaker apparatus **1** may include a main housing **100** (see FIG. **4**). The main housing **100** may be arranged on a lower portion of the inner housing **50** to cover the inner housing **50**. The inner housing **50** may be arranged to be seated on the main housing **100**. In other words, the main housing **100** may be configured to accommodate internal components (e.g., a processor, diaphragms, circuits etc.) of the speaker apparatus **1**, such as the speaker **70**, by covering the inner housing **50** as a lower cover.

[0067] The speaker apparatus **1** may include a sub-housing **200** (see FIG. **5**). The sub-housing **200** may be configured to be coupled with the main housing **100**. A detailed description of the coupling between the sub-housing **200** and the main housing **100** will be described later.

[0068] The speaker apparatus **1** may include a connection hole **101** (see FIG. **2**) through which wires from other external home appliances or an external power source for power supply may be physically connected. More particularly, the connection hole **101** may be provided on the rear cover **40** of the main housing **100**. Through the connection hole **101**, the speaker apparatus **1** may be connected to the outside and receive electrical signals to generate sound. However, the speaker apparatus **1** is not limited to such a method, and may be wirelessly connected to other home appliances via a wireless communication method to transmit and receive signals.

[0069] The speaker apparatus **1** may include a plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e*. The main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may be categorized as a cone speaker, a dome speaker, a flat panel speaker, a ribbon speaker, a horn speaker, and a micro speaker depending on the shape of the vibration plate **71***a*, **71***b*, **71***c*, **71***d* and **71***e*.

[0070] The cone speaker may have a cone-shaped vibration plate, which has generally flat frequency characteristic and allows sound in the low-frequency range to spread across the entire 360-degree direction.

[0071] The dome speaker may be a speaker with a dome-shaped vibration plate, and may be characterized in that the reproduction range is wide and spread evenly.

[0072] The flat panel speaker may refer to a speaker with a flat vibration plate without curvature, so that there is no time difference in the arrival of the sound generated from a plurality of speakers. The generated sound may have the characteristics of a spherical wave and a plane wave rather than a pure spherical wave, which may be characterized in that the sound may be sent far away.

[0073] The ribbon speaker may be a speaker with a ribbon-like shape, which has the characteristic that the driving force for moving the vibration plate is not generated from the voice coil, but is directly generated from the vibration plate itself.

[0074] The horn speaker may be a speaker having a horn-like shape, and may have a characteristic of concentrating sound in one direction.

[0075] The micro speaker may refer to a speaker with a size of 40-50 mm or less, and may be used mainly for small-sized products.

[0076] The plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may be used as flat panel speakers, but are not limited thereto.

[0077] The description of the types of speakers mentioned above may be equally applied to the front speakers **80***b* and **80***c*.

[0078] The main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may be arranged on side walls of the inner housing **50** and function as side speakers.

[0079] The main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e*each installed on the side walls of the inner housing **50** may be configured to generate sound in a left-and-right direction of the speaker apparatus **1**.

[0080] The speaker may include a plurality of front speakers **80***b* and **80***c* mounted on a front wall **60** of the inner housing **50**. The front speakers **80***b* and **80***c* may be configured as two front speakers. The front speakers **80***b* and **80***c* may include a first front speaker **80***b* arranged on a right side of the front wall **60**, and a second front speaker **80***c* arranged on a left side of the front wall **60**.

[0081] The first front speaker **80***b* may generate sound toward a front right side of the inner housing **50**, and the second front speaker **80***c* may generate sound toward a front left side of the inner housing **50**. However, the shape, position, and number of the front speakers **80***b* and **80***c* are not limited thereto.

[0082] The plurality of front speakers **80***b* and **80***c* may be mounted on the inner housing **50**, respectively. More particularly, the front wall **60** of the inner housing **50** may include front speaker holes **80***b*' and **80***c*' arranged corresponding to the plurality of front speakers **80***b* and **80***c* for mounting the plurality of front speakers **80***b* and **80***c*. The plurality of front speakers **80***b* and **80***c* may be mounted on the inner housing **50** by being inserted into the front speaker holes **80***b*' and **80***c*', respectively.

[0083] The plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may be mounted on the front wall **60** of the inner housing **50**. Five main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may be provided. The plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may include a first main speaker **70***a* and a second main speaker **70***b* arranged on the right side of the front wall **60**. The plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may include a fourth main speaker **70***d* and a fifth main speaker **70***e* arranged on the left side of the front wall **60**. The plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may include a third main speaker **70***c* arranged at the center of the front wall **60**. However, the shape, position, and number of the main speakers **70** are not limited thereto and may be provided in various ways.

[0084] The front wall **60** of the inner housing **50** may be provided with main speaker holes **70***a*', **70***b*', **70***c*', **70***d*' and **70***e*' for mounting the plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e*. The plurality of main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e* may be mounted on the inner housing **50** by being inserted into the main speaker holes **70***a*', **70***b*', **70***c*', **70***d*' and **70***e*', respectively.

[0085] The speaker apparatus **1** may include a plurality of woofers. The woofers may be components for generating low-frequency sounds of 3,000 Hz or less in the audible frequency range. The woofers may be included in the speaker apparatus **1** to improve consumer satisfaction by generating sound with a range of different frequencies.

[0086] The woofers may be installed along the front wall **60** of the inner housing **50**. The woofers may be installed along the main housing **100**. However, the installation position of the woofers mentioned above is merely by way of examples, and it should be understood that the woofers may be arranged at various positions on the speaker **1**.

[0087] The inner housing **50** may include woofer holes on which the woofers may be mounted. The woofers may be mounted on the inner housing **50** by being inserted into the woofer holes, respectively.

[0088] Meanwhile, the arrangement and number of the plurality of main speakers **70**, the front speakers **80***b* and **80***c*, and the woofers, as well as the main speaker holes **70***a*, **70***b*, **70***c*, **70***d* and **70***e*, the front speaker holes **80***b*', **80***c*', and the woofer holes for accommodating them, are merely by way of examples, and may be provided in various ways, as needed.

[0089] The front cover **30** may be attached to a front surface of the inner housing **50**. The front cover **30** may be coupled to the front wall **60** to protect the front speakers **80***b* and **80***c*, the main speakers **70**, and the plurality of woofers arranged on the front wall **60**. The front cover **30** may be provided in the shape of a grille. According to such a shape, the front cover **30** may prevent the front speakers **80***b* and **80***c*, the main speakers **70***a*, **70***b*, **70***c*, **70***d* and **70***e*, and the woofers from being pressed by pointed members, and may also have an aesthetic effect.

[0090] The speaker apparatus **1** may include an operating portion **21**. The operating portion **21** may be provided with an on/off button and a volume control button for the speaker apparatus **1** and allow a user to easily control the speaker apparatus **1**.

[0091] The operating portion **21** may be arranged to be installable on the inner housing **50**. The inner housing **50** may include an inner housing hole **21'** in which the operating portion **21** is installed. The operating portion **21** may be arranged to be inserted into the inner housing hole **21'**.

[0092] The upper cover **31** may include an upper cover hole **31'** that allows the operating portion **21** installed in the inner housing **50** to be exposed to the outside. The upper cover hole **31'** may be configured to pass through the upper cover **31**. The upper cover hole **31'** may be provided in the same shape as the operating portion **21**.

[0093] FIG. **4** is a view illustrating an interior of the speaker apparatus according to one or more embodiments.

FIG. **5** is a view illustrating a state in which the sub-housing is fastened to the main housing according to one or more embodiments.

[0094] Referring to FIGS. **4** to **5**, the speaker apparatus **1** may include an electronic component **91** arranged therein. More particularly, the electronic component **91** may be arranged to be seated on the main housing **100**. The electronic component **91** may be provided as a printed board assembly (PBA) or the like connected to the plurality of front speakers **80***b* and **80***c*, the main speakers **70**, and the woofers.

[0095] The electronic component **91** may include a printed circuit board (PCB). The PCB may be provided as a board, such as a substrate, on which conductive circuitry is formed on or within an insulating base for electrically connecting components. The PCB may be configured to electrically contact the components of the speaker apparatus **1** described above to control the components thereof.

[0096] The speaker apparatus **1** may include an electronic component cover **92**. The electronic component cover **92** may be arranged to be disposed between the inner housing **50** and the electronic component **91**. The electronic component cover **92** may be configured to cover the electronic component **91** to protect the electronic component **91**.

[0097] The speaker apparatus **1** may include a light emitting lamp board **93**. The light emitting lamp board **93** may be in electrical contact with the electronic component **91**. The electronic component **91** may control a light emitting lamp on the light emitting lamp board **93** to emit light.

[0098] The light emitting lamp board **93** may be disposed adjacent to the sub-housing **200**. The light emitting lamp board **93** may be provided in a shape corresponding to the sub-housing **200**. The sub-housing **200** may include a light-transmissive material through which light is transmitted. Thus, the light emitted by the light emitting lamp may be emitted to the outside through the sub-housing **200**, and an aesthetic effect due to light may be generated when the speaker apparatus **1** is operated.

[0099] FIG. **6** is a view illustrating only the main housing and the sub-housing separated from the speaker apparatus **1** according to one or more embodiments. FIG. **7** is an enlarged view illustrating area A of FIG. **6**. FIG. **8** is a side cross-sectional view after cutting the main housing and the sub-housing of FIG. **7** along the cutting surface a-a'. FIG. **9** is a view illustrating a state in which the sub-housing is fastened to the main housing while a deformation portion of FIG. **8** elastically deforms.

[0100] Referring to FIGS. **6** to **9**, the main housing **100** may include an elastic hole **110**. The main housing **100** may include a deformation portion **132** arranged along one side of the elastic hole **110**. The main housing **100** may include a body portion **131** arranged along the other side of the elastic hole **110**. In other words, the elastic hole **110** may be formed with the deformation portion **132** on one side and the body portion **131** on the other side.

[0101] A width D of the deformation portion **132** may be formed narrower than a width d of the body portion **131**. Accordingly, the deformation portion **132** may be configured to elastically deform in response to coupling of the sub-housing **200** to the main housing **100**.

[0102] The sub-housing **200** may be accommodated in the elastic hole **110** and coupled to the main housing **100**. The elastic hole **110** may be sized corresponding to the sub-housing **200** to accommodate the sub-housing **200**. The elastic hole **110** may be configured to extend along a longitudinal direction of the main housing **100**.

[0103] The main housing **100** may include a connection rib **120** arranged across the elastic hole **110** to connect and support the deformation portion **132** with the body portion **131**. Accordingly, the elastic hole **110** may include a first elastic hole **110***a* and a second elastic hole **110***b*, which are divided by the connection rib **120**.

[0104] The connection ribs **120** may be provided in a plurality. The plurality of connection ribs **120***a* and **120***b* may be arranged to be spaced apart from each other to divide the elastic holes **110***a* and **110***b*. The plurality of connection ribs **120***a* and **120***b* may be arranged to be equally spaced apart from each other. The plurality of connection ribs **120***a* and **120***b* may extend along a direction in which the elastic hole **110** extends. Here, the interval at which the plurality of connection ribs **120***a* and **120***b* are spaced apart from each other may be varied depending on the shape of the speaker apparatus **1**.

[0105] The deformation portion **132** may be elastically deformable to move the sub-housing **200** away from the body portion **131** when the sub-housing **200** is coupled to the main housing **100**. The deformation portion **132** may be elastically deformed based on the point where the deformation portion **132** is supported in contact with the connection rib **120**. In other words, a portion of the deformation portion **132** that is in contact with the plurality of connection ribs **120***a* and **120***b* may not be elastically deformed when the sub-housing **200** is coupled to the main housing **100**. Accordingly, the deformation portion **132** may include a first deformation portion **132***a* and a second deformation portion **132***b*, which are differentiated based on the point of contact with the connection rib **120***b*.

[0106] The first deformation portion **132***a* may be arranged to correspond to the first elastic hole **110***a*. The second deformation portion **132***b* may be arranged to correspond to the second elastic hole **110***b*. The first deformation portion **132***a* and the second deformation portion **132***b* may be elastically deformable, respectively, when the sub-housing **200** is coupled to the main housing **100**.

[0107] The amount of elastic deformation of each of the deformation portions **132***a* and **132***b* may be controlled by a distance at which the plurality of connection ribs **120***a* and **120***b* are spaced apart from each other. In other words, since the portion of the deformation portion **132** that is in contact with the connection rib **120** does not deform when the main housing **100** and the sub-housing **200** are coupled, the shorter the distance at which the plurality of connection ribs **120** are spaced apart from each other, the less the amount of deformation of the elastic portion may be.

[0108] The sub-housing **200** may be coupled to the main housing **210** via a snap fit mechanism that employs a flexible feature to snap into place itself or allow a corresponding part to snap into engagement. The snap fit mechanism may include a protruding element, such as a hook, bead, or ridge, configured to engage with a corresponding hole, undercut, or groove in a mating part. In some embodiments, the protruding element (e.g., a hook **210**) is positioned on the sub-housing **200**, while an elastic hole **110** is formed in the main housing **210**. However, the present disclosure is not limited to this configuration; alternatively, the protruding element may be located on the main housing **210**, with the elastic hole **110** formed in the sub-housing **200**. Additionally, in certain embodiments, the main housing **210** provides the flexible feature via the hole, although either or both of the main housing **210** and the sub-housing **200** may incorporate a flexible feature to implement the snap fit mechanism.

[0109] In one or more embodiments, the sub-housing **200** may include a hook **210** inserted into the elastic hole **110** to support the sub-housing **200** on the main housing **100**. The sub-housing **200** may be coupled to the main housing **100** by hooking the hook **210** arranged in the sub-housing **200** to the main housing **100**. More particularly, the main housing **100** may include a hook fastening groove **112**. The hook **210** may be fastened into the hook fastening groove **112** of the main housing **100** to couple the sub-housing **200** to the main housing **100**. The hook fastening groove **112** may include a plurality of hook fastening grooves **112***a* and **112***b*.

[0110] The hook **210** of the sub-housing **200** may be inserted into the elastic hole **110** of the main housing **100**. When the hook **210** is inserted into the elastic hole **110** and the sub-housing **200** is coupled to the main housing **100**, the deformation portion **132** may be elastically deformed.

[0111] Since the hook **210** does not elastically deform when the main housing **100** and the sub-housing **200** are coupled, the hook **210** may not include a length for elastically deforming. Since the hook **210** does not elastically deform, damage to the hook **210** may be prevented. Furthermore, since the hook **210** does not need to include a length for elastic deformation, an internal space of the speaker apparatus **1** may be utilized to the extent of the shortened length. In other words, a space for positioning the light emitting lamp board **93**, which is disposed adjacent to the sub-housing **200**, may be secured.

[0112] A plurality of hooks **210***a*, **210***b* of the sub-housing **200** may be provided. The plurality of hooks **210***a*, **210***b* may include a first hook **210***a* inserted into the first elastic hole **110***a*. The plurality of hooks **210***a*, **210***b* may include a second hook **210***b* inserted into the second elastic hole **110***b*.

[0113] The sub-housing **200** may further include a guide rib **220** that guides the sub-housing **200** being received in the elastic hole **110**. The guide rib **220** may be formed in the sub-housing **200** at a position corresponding to a point where the hook **210** is positioned. The guide rib **220** may extend in a direction corresponding to the direction in which the hook **210** extends from the sub-housing **200**. The guide rib **220** may include a plurality of guide ribs **220***a*, **220***b*.

[0114] The main housing **100** may include a side wall **111** protruding from the deformation portion **132** so as to define one side of the elastic hole **110**.

[0115] The side wall **111** may be configured to protrude from the deformation portion **132** toward the body portion **131**. The side wall **111** formed on the deformation portion **132** may be configured to protrude from the connection rib **120**. The side wall **111** may include a plurality of side walls **111***a*, **111***b*.

[0116] The main housing **100** may include a guide rib groove **113**. More particularly, the guide rib groove **113** may be formed in the side wall **111**. When the guide rib **220** is inserted into the elastic hole **110**, the guide rib **220** may be configured to be seated in the guide rib groove **113**. The guide rib groove **113** may include guide rib grooves **113***a*, **113***b*.

[0117] FIG. **10** is a view illustrating a state in which the sub-housing is fastened to the main housing while a first deformation portion and a second deformation portion according to one or more embodiments are elastically deformed, respectively. FIG. **11** is a view illustrating a state in which the sub-housing is fastened to the main housing according to one or more embodiments. FIG. **12** is a view illustrating FIG. **11** from another angle.

[0118] Referring to FIGS. **10** to **12**, the sub-housing **200** may be coupled to cover the main housing **100**. The detailed process by which the sub-housing **200** is coupled to the main housing **100** will be described below.

[0119] The guide rib **220** of the sub-housing **200** may be inserted into the elastic hole **110**. The guide rib **220** inserted into the elastic hole **110** may be seated in the guide rib groove **113** arranged in the deformation portion **132** of the main housing **100**. Thereafter, when the hook **210** is inserted into the elastic hole **110**, the guide rib **220** may press against the guide rib groove **113**. As the guide rib groove **113** is pressed, the deformation portion **132** of the main housing **100** may be pressed.

[0120] In response to the guide rib **220** pressing the deformation portion **132**, the deformation portion **132** may elastically deform in the direction away from the body portion **131**. As the deformation portion **132** elastically deforms, the size of the elastic hole **110** may increase. As the size of the elastic hole **110** increases, the hook **210** may be inserted into the elastic hole **110**. The hook **210** inserted into the elastic hole **110** may be seated in the hook fastening groove **112** of the body portion **131**. After the hook **210** is seated in the hook fastening groove **112**, the deformation portion **132** may be elastically deformed in the direction closer to the body portion **131**. As a result, the sub-housing **200** may be coupled to the main housing **100** to cover the elastic hole **110** of the main housing **100**.

[0121] While the present disclosure has been particularly described with reference to exemplary embodiments, it should be understood by those of skilled in the art that various changes in form and details may be made without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An electronic apparatus comprising:

a main housing configured to accommodate electronic components and comprising an elastic hole; and

a sub-housing configured to be coupled to the main housing through the elastic hole;

wherein the main housing comprises:

a deformation portion provided along one side of the elastic hole,

a body portion provided along another side of the elastic hole, and

a connection rib provided across the elastic hole to connect and support the deformation portion and the body portion,

wherein the sub-housing comprises a hook inserted into the elastic hole to couple the sub-housing to the main housing, and

during coupling of the sub-housing to the main housing, the deformation portion undergoes elastic deformation to enable the hook to be inserted into the elastic hole.

2. The electronic apparatus of claim **1**, wherein

the elastic hole comprises a first elastic hole and a second elastic hole separated by the connection rib, and

the hook comprises a first hook inserted into the first elastic hole and a second hook inserted into the second elastic hole.

3. The electronic apparatus of claim **2**, wherein

the deformation portion comprises a first deformation portion arranged to correspond to the first elastic hole and a second deformation portion arranged to correspond to the second elastic hole, and

the first deformation portion and the second deformation portion are configured to elastically deform, respectively, during the coupling of the sub-housing to the main housing.

4. The electronic apparatus of claim **1**, wherein

the sub-housing further comprises a guide rib configured to guide reception of the sub-housing into the elastic hole, and

the guide rib is configured to press the deformation portion of the main housing during the coupling of the sub-housing to the main housing.

5. The electronic apparatus of claim **4**, wherein the main housing further comprises a side wall protruding from the deformation portion to define one side of the elastic hole, and a guide rib groove provided in the side wall to allow the guide rib to be seated therein.

6. The electronic apparatus of claim **5**, wherein

the main housing further comprises a hook fastening groove to which the hook is fastened, and

the hook fastening groove is at a position corresponding to the guide rib groove.

7. The electronic apparatus of claim **4**, wherein the guide rib extends in a direction corresponding to a direction in which the hook extends from the sub-housing.

8. The electronic apparatus of claim **1**, wherein the elastic hole and the deformation portion extend along a longitudinal direction of the main housing.

9. The electronic apparatus of claim **8**, wherein

the connection rib comprises a plurality of connection ribs provided along a direction in which the elastic hole extends, and

the plurality of connection ribs are spaced apart from each other to partition the elastic hole.

10. The electronic apparatus of claim **1**, wherein the sub-housing is coupled to the main housing to cover the elastic hole.

11. The electronic apparatus of claim **2**, wherein a width of the deformation portion is narrower than a width of the body portion.

12. The electronic apparatus of claim **1**, wherein the sub-housing comprises a light-transmissive material.

13. The electronic apparatus of claim **1**, wherein the electronic apparatus is a speaker apparatus.

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